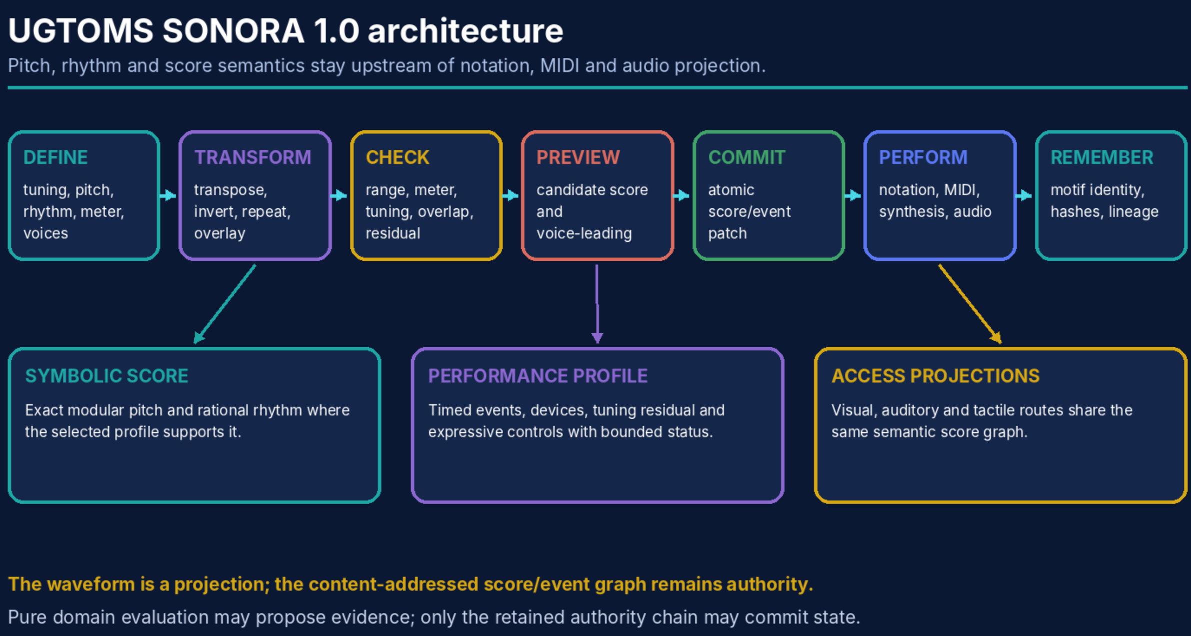
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

# UGTOMS SONORA 1.0

## Pitch-Class, Rhythm, Voice-Leading and Executable Notation Cartridge




`ugtoms.domain.sonora@1.0.0`

| | |
|---|---|
| **Component** | SONORA |
| **Canonical identity** | ugtoms.domain.sonora@1.0.0 |
| **Version / date** | 1.0.0 / 30 August 2026 |
| **Document type** | Domain cartridge specification and implementation-ready profile |
| **Status** | Formal domain cartridge; specification self-checked; musical, cultural and interchange completeness not claimed |

**ATTRIBUTION AND EVIDENCE BOUNDARY**

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

# Contents and Reading Rule



**READING RULE**

The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.

# Executive Decision and Release Scope

**FORMAL DECISION**

GO for SONORA as the second cartridge. Music is represented as a typed score-and-performance graph with explicit tuning, timing, voice, notation and projection profiles. No single tuning system, notation tradition or harmonic analysis is promoted as universal.

SONORA makes the Foundation 5.0 executable-notation idea immediately tangible: a symbol may resolve literal identity, canonical geometry and a typed musical operation while those faces remain separate. The core result is not a synthesizer. It is an exact-or-bounded event substrate that can feed many synthesizers, notation systems, games and learning routes.

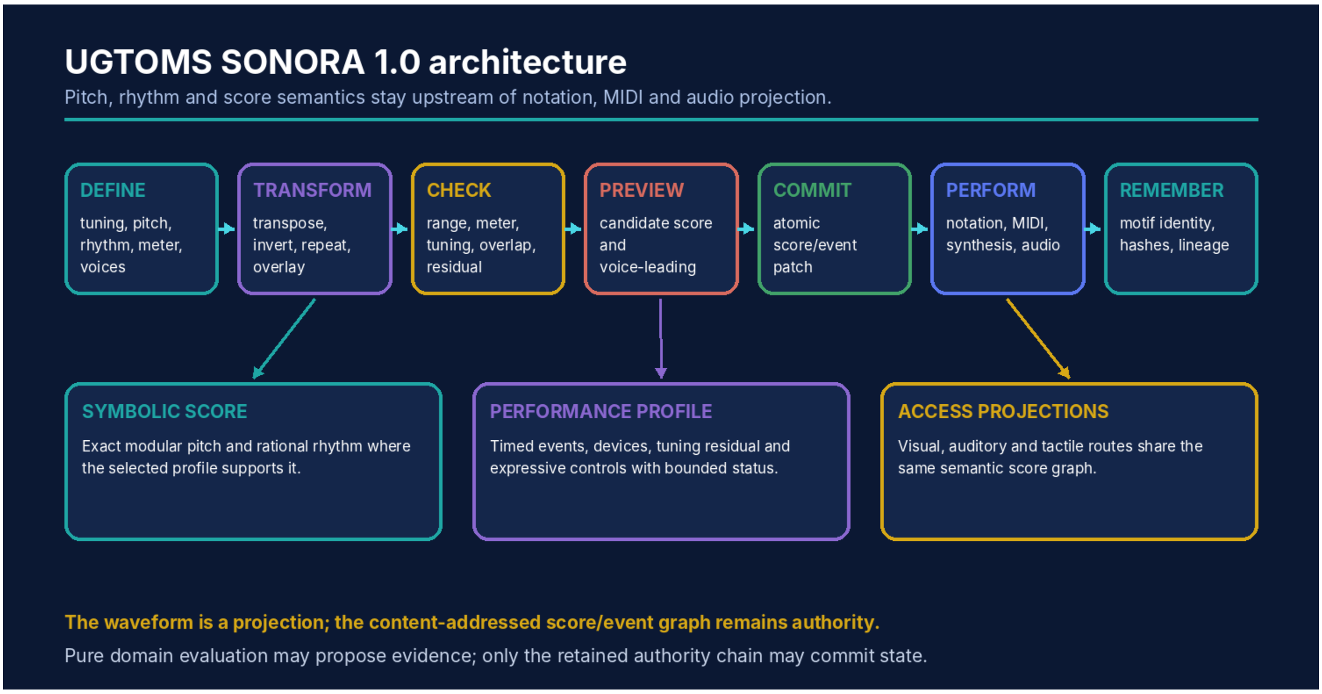



*Figure 1. SONORA keeps score semantics authoritative and performance/audio downstream.*

# Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

**ENGINEERING-DERIVED DOMAIN CONTENT**

The pitch, rhythm, tuning, notation, voice-leading and music-event definitions are engineering-derived. No external music-theory corpus, notation standard, MIDI/MusicXML specification, instrument database, cultural canon or copyrighted musical work is imported by this PDF.

| ID | Registered source | Use in this cartridge | Pages |
|---|---|---|---|
| S1 | UGTS__FOUNDATION__UNICODE_v5.pdf | Foundation 5.0 OperatorCells, exactness/capability, Unicode literal/glyph/semantic separation, hot codebooks and authority boundary. | pp. 1-21 |
| S2 | UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf | Typed OperatorSpec, domains/codomains, units, laws, effects, dependencies, order profiles, certificates and open extension contracts. | pp. 2-18 |
| S3 | UGTS_KC_3_6_Tom_Klootwijk.pdf | Definitions as first-class content-addressed data, resolvable dependencies and explicit operation order. | pp. 1-14 |
| S4 | Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf | Canonical support → compatibility → guard → event → transition → lineage sequence; packed-state and performance evidence discipline. | pp. 2-15 |
| S5 | UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf | Pattern, field, topology, kinematics, bounded dynamics and hybrid-event architecture. | pp. 3-15 |
| S6 | UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf | Stable identity, proposal verification, deterministic commit, checkpoints, replay and downstream presentation. | pp. 3-16 |
| S7 | UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf | Versioned projects, deterministic game-world snapshots/hashes, compact records and self-contained delivery. | pp. 2-16 |
| S8 | UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf | Observation versus authority, uncertainty, provenance, atomic patches, checkpoints and evidence-bearing lineage. | pp. 3-14 |
| S9 | UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf | Finite packing, log-polar charts, interval guards, winding and explicit topology/rotation distinctions. | pp. 1-15 |
| S10 | UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf | Component-scoped semantic identities and explicit release graphs instead of one misleading global number ladder. | pp. 1-6 |

# Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_SONORA:

```
D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)
```

| Field | Contract |
|---|---|
| T | Entity, value and state types |
| O | Typed operators and transformations |
| R | Relations, graphs, topology and converse pairs |
| U | Units, dimensions, coordinate and profile conventions |
| I | Invariants, conservation laws and constraints |
| Q | Dependency, precedence, phase and reduction order |
| E | Observations, evidence, uncertainty and provenance |
| G | Guards, thresholds and admission predicates |
| Delta | Proposed and committed state patches |
| L | Lineage, replay, history and migration |
| Cert | Exactness, residual, interval, scope and status certificates |
| K | Cold atlas, hot codebook and packed representation |
| Pi | Optional visual, audio, physical or device projection |

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

# General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
                laws, exactness, effects, capabilities, dependencies,
                order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

**EXAMPLE RECORD**

music.transform.transpose consumes a PitchClass or ScoreFragment under a TuningProfile and integer/interval parameter. Its exactness is modular-exact for the declared Z_n profile. A rendered glyph such as a sharp sign does not override the tuning or spelling policy.

## Normative phase schedule

| Phase | Meaning |
|---|---|
| 1-3 | parse, normalize and resolve content-addressed references |
| 4-6 | type, shape and unit validation; dependency closure |
| 7-9 | construct candidates; apply declared composition and reduction order |
| 10-12 | evaluate exact or bounded results; issue capability and residual certificate |

| Phase | Meaning |
|---|---|
| 13 | support, compatibility and guard classification |
| 14 | verified proposal and deterministic conflict resolution |
| 15 | atomic commit, lineage, replay and optional projection |

# Formal SONORA State

```
q_SONORA = (P, T, R, M, V, H, A, X, E, L)

P pitch/frequency space   T tuning/temperament   R rhythm/time map
M meter/form              V voices/instruments   H harmonic/motif graph
A articulation/dynamics   X candidate/committed events
E performance evidence    L lineage/replay
```

| Object | Authority role | Boundary |
|---|---|---|
| TuningProfile | Maps symbols/classes to frequencies and defines equivalence assumptions. | 12-TET is one profile, not the default truth for all music. |
| ScoreEvent | Stable event identity with onset, duration, pitch/rest, voice and controls. | A drawn notehead is a view, not event identity. |
| ScoreGraph | Ordered/simultaneous/tied/phrased relations plus dependencies. | Equal coordinates or pitch classes do not collapse voices or lineage. |
| PerformanceEvent | Time-mapped device/output event with residual/status. | Performance may vary without rewriting score authority. |
| AudioEvidence | Rendered or recorded waveform and device/room profile. | Waveform is downstream evidence/projection, not the complete score. |

**THREE DISTINCT IDENTITIES**

Literal spelling (for example C-sharp), pitch-class identity and acoustic frequency may coincide under one profile and differ under another. SONORA records the maps rather than declaring them identical.

# Pitch, Tuning and Exact Transformations

```
For a declared n-class profile:
    pitch class p in Z_n
    T_k(p) = p + k mod n
    I_k(p) = k - p mod n
    directed_interval(p,q) = q - p mod n
```

These equations are exact within the selected modular profile. Absolute pitch, cents/log-frequency, spelling and register remain separate typed faces. Microtonal, just-intonation, non-octave and adaptive tuning profiles can implement the same ABI with different maps and certificates.

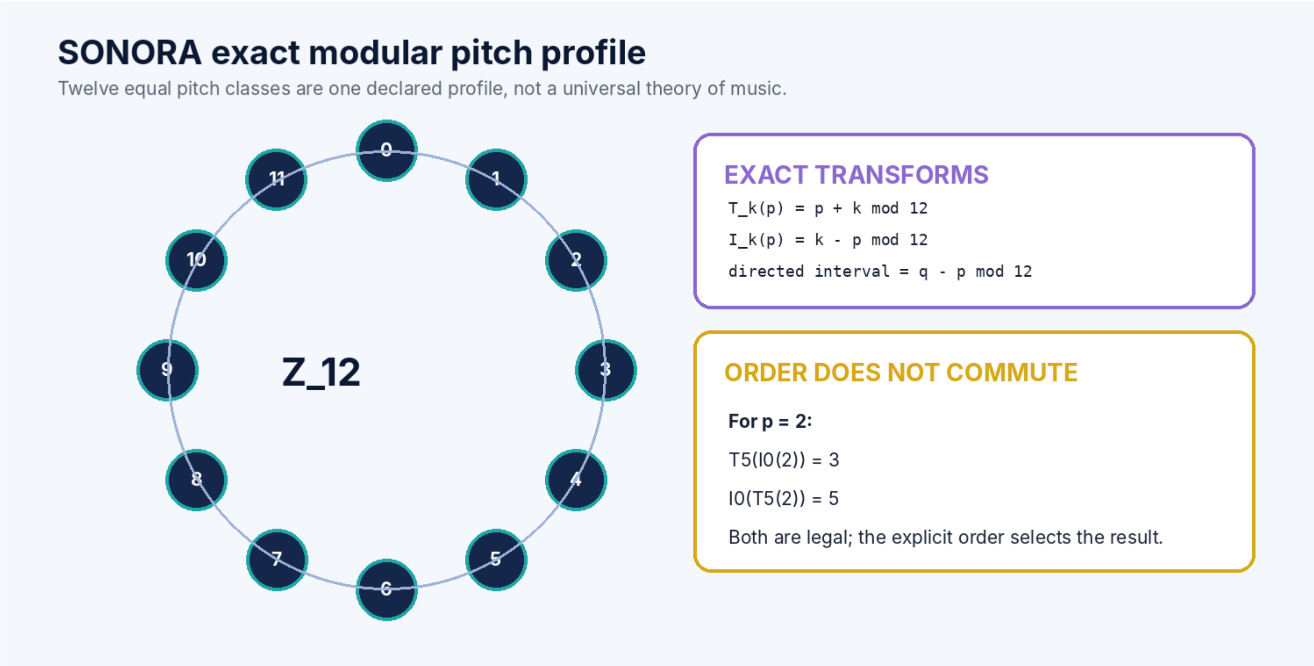



*Figure 2. A 12-class circle demonstrates one exact modular profile and a noncommuting order example.*

## Pitch capability ladder

| Capability | Meaning |
|---|---|
| literal-only | Notation spelling is stored without a pitch-resolution claim. |
| class-exact | Exact arithmetic in a declared finite pitch-class group. |
| frequency-mapped | Symbols/classes map to numeric frequencies under a tuning profile. |
| instrument-bounded | Range, fingering/device or intonation limits are declared. |
| performance-measured | Observed/recorded frequency and timing residuals are available. |

# Rhythm, Meter and Event Authority

```
ScoreEvent = (id, onset, duration, pitch_or_rest, voice, velocity_or_dynamic,
              articulation, dependencies, profile, provenance, content_hash)
```

Onset and duration may be exact rationals in score time. Tempo maps convert score time to performance time. Swing, rubato, fermata, humanization and latency are typed transforms or evidence profiles; they never overwrite the exact symbolic source silently.

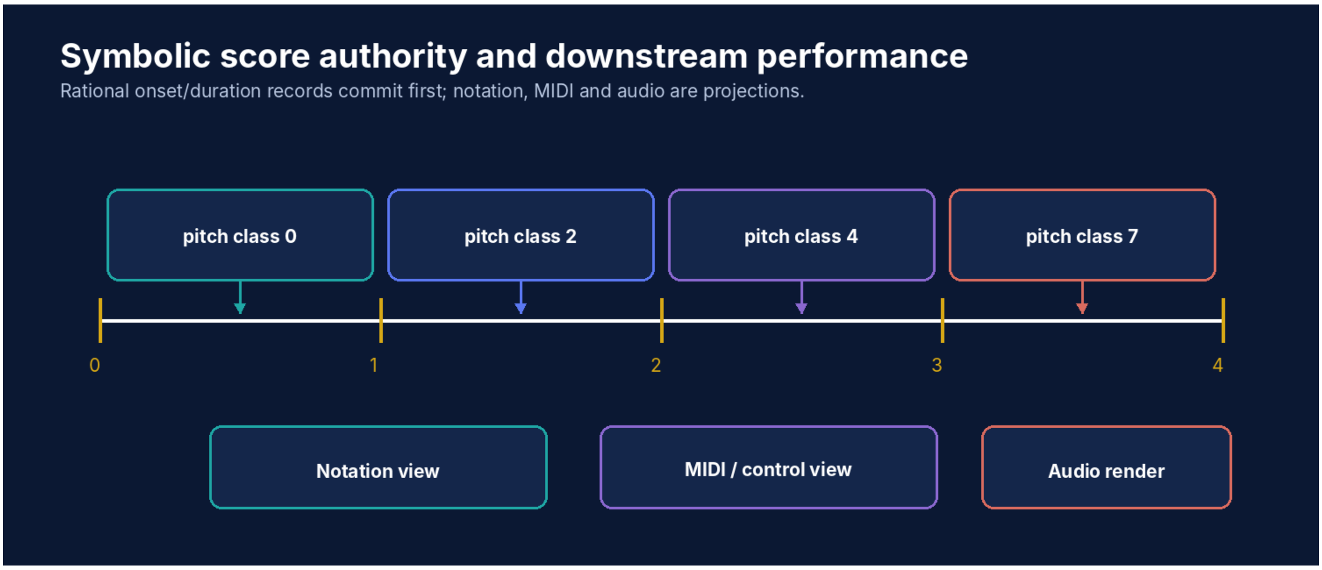



*Figure 3. Exact score events commit before notation, MIDI and audio projection.*

## Verified event predicate

```
verified(e,q) = voice_exists
              and tuning_resolves
              and range_ok
              and meter_ok
              and overlap_policy_ok
              and dependency_order_ok
              and quantization_residual <= margin
```

## Meter and conflict semantics

| Case | Deterministic treatment |
|---|---|
| measure duration mismatch | reject or retain an explicitly open/cadenza profile |
| two events in a monophonic voice overlap | reject, tie, truncate or route by declared conflict policy |
| same onset in different voices | allowed when polyphony and channel/device guards pass |
| tempo change at an event boundary | evaluate under explicit phase/order rule |
| late device event | record performance residual; do not move the canonical score event silently |

# Voices, Harmony and Operation Order

A chord is a pitch multiset plus voicing, register, spelling and event identity. Harmonic labels are optional profile metadata; they do not become universal truth conditions.

| Operator composition | Result-affecting distinction |
|---|---|
| transpose then invert vs invert then transpose | both legal modular maps can produce different results |
| retrograde then augment vs augment then retrograde | event order and transformed boundary positions differ when tempo/meter profiles are scoped |
| overlay then quantize vs quantize each voice then overlay | conflict and residual outcomes may differ |
| voice-lead then range-clamp vs range-filter then voice-lead | candidate space and optimal mapping differ |

## Voice-leading proposal

```
VoiceLeadProposal = (source_chord, target_chord, voice_map, cost_profile,
                     range_guards, crossing_policy, retained_common_tones, status)
```

**ANALYSIS BOUNDARY**

A minimum-cost voice-leading map is exact only relative to its declared voices, cost function, admissible mappings and constraints. It is not a theorem about artistic quality or historical practice.

# Executable Notation OperatorCells

| Literal / fallback | Typed face | What is not inferred |
|---|---|---|
| # / sharp | spelling or pitch-alteration operator under a tuning/notation profile | universal frequency ratio or enharmonic identity |
| b / flat | spelling or pitch-alteration operator under a profile | semantic inverse of sharp in every context |
| natural | cancellation/reset operator scoped to notation context | a geometric mirror relation |
| tie | continuation relation between compatible events | mere equality of pitch or adjacency |
| repeat | control-flow relation with count/ending/profile | opaque copying of audio samples |
| crescendo/ decrescendo | time-scoped dynamic transform | a fixed acoustic amplitude law |
| rest | explicit time-bearing silent event | missing data or muted output automatically |

**CONVERSE AND GLYPH RULE**

Sharp and flat are not registered as a Foundation-style direct/converse pair by default. Their surface shapes, historical spellings, tuning effects and cancellation behavior are profile-dependent. A visual reflection or bit flip never supplies musical inverse semantics.

## Notation profile layers

- Literal identity: exact Unicode/ASCII spelling and normalization policy.
- Semantic identity: typed alteration, relation, articulation or control operator.
- Glyph profile: canonical shape or font-exact rendering metadata.
- Tuning profile: mapping from spelling/class to frequency or ratio.
- Performance profile: device/output interpretation and residual status.

# Deterministic Demonstration

```
profile: Z_12, 4/4 meter
motif pitch classes: [0, 2, 4, 7]
durations: [1/4, 1/4, 1/4, 1/4]
T_5(motif): [5, 7, 9, 0]
I_0(motif): [0, 10, 8, 5]
measure total: 1 whole note -> PASS
```

| Check | Result | Certificate / reason |
| --- | --- | --- |
| four quarter durations in 4/4 | ACCEPT | meter-exact |
| T5(I0(2)) | 3 | order trace: invert → transpose |
| I0(T5(2)) | 5 | order trace: transpose → invert |
| voice range [0,8], transformed note 9 | REJECT | range-exceeded |
| same-onset notes in two declared polyphonic voices | ACCEPT | polyphony-profile-pass |
| audio render differs by synthesizer | ACCEPT projection difference | score hash unchanged |

## Score and performance queries

- events_in_measure(measure_id) → ordered exact score events
- interval_path(voice_id) → directed intervals plus spelling/tuning profile
- motif_occurrences(motif_id, transforms_allowed) → lineage-preserving references
- performance_residual(score_event_id) → timing/pitch/device status
- explain_rejection(proposal_id) → first failed range, meter, tuning, overlap or dependency gate

# Performance, Audio and Accessibility

| Layer | Authority and evidence rule |
|---|---|
| Notation view | Reconstructs the score graph; layout choices do not change event identity. |
| MIDI/control stream | Adapter mapping with channel, resolution and feature-loss certificate. |
| Synthesis/sample engine | Downstream audio projection; model/sample/version is recorded. |
| Recorded performance | Observation with microphone/device/room/time profile and uncertainty. |
| Tactile score | Semantic event/relationship route using shape, position and texture; color is optional. |
| Sonification of non-audio state | Projection profile must declare mapping and avoid presenting metaphor as exact identity. |

**ACCESS PARITY**

A visual staff, spoken event list, tactile timeline and keyboard/gamepad editor may all operate on the same score graph. Parity requires the same objects, relationships, available actions, verification results, commit consequences and replay lineage.

## Compression and reconstructibility

- Motifs can be stored once and referenced with transposition, inversion, time scale and voice maps.
- Rational durations and modular pitch classes pack exactly inside their declared ranges.
- Human timing, timbre, room response and expressive detail cannot be reconstructed from a seed unless they are generated by the declared profile.
- A hot opcode remains a local alias bound to the notation/operator atlas hash.

## External adapters

| Adapter | Status in 1.0 |
|---|---|
| MIDI | specified boundary; complete standard conformance not claimed |
| MusicXML or other notation exchange | specified boundary; profile/version and loss report required |
| Audio plugin/synthesizer | replaceable projection with deterministic seed option and device evidence |
| Instrument database | external source for ranges, techniques and transposition; no built-in universal table |
| Cultural/theory profile | open module with its own provenance, terminology and non-claims |

# Reference Validation, Boundaries and Kill Criteria

**REFERENCE SELF-CHECK RESULT**

24/24 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

| Checked area | Result |
|---|---|
| Catalog integrity | 48 unique SNR IDs; required fields present. |
| Modular arithmetic | Transposition/inversion fixtures reproduce exact Z_12 results. |
| Order trace | Noncommuting transform example produces two distinct certified results. |
| Rational rhythm | 4/4 fixture totals exactly; mismatch fixture rejects. |
| Voice guards | Range, polyphony and overlap reason codes are deterministic. |
| Projection boundary | Changing audio renderer leaves canonical score hash unchanged. |

## Explicit non-claims

- No tuning, notation, harmonic theory, performance practice or cultural framework is claimed universal.
- No copyrighted composition, proprietary sample, instrument database or external standard is reproduced.
- No audio rendering is claimed perceptually transparent or equivalent across devices.
- No aesthetic, educational, therapeutic or accessibility outcome is independently validated.
- No performance/compression advantage is claimed without an implementation and equal-quality benchmark.

## Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

# Complete SONORA 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

| ID | Family | Mechanism | Core contract |
|---|---|---|---|
| SNR001 | Pitch | Tuning profile | Declares frequency mapping, reference pitch and temperament assumptions. |
| SNR002 | Pitch | Absolute pitch | Frequency or log-frequency value with unit and uncertainty. |
| SNR003 | Pitch | Pitch class | Modular class in Z_n under a selected tuning profile. |
| SNR004 | Pitch | Pitch spelling | Literal notation identity remains distinct from acoustic pitch equivalence. |
| SNR005 | Pitch | Instrument range | Playable/support region for a voice or device profile. |
| SNR006 | Pitch | Temperament map | Maps symbolic classes to frequencies under a declared profile. |
| SNR007 | Transform | Directed interval | q - p modulo n, with direction preserved. |
| SNR008 | Transform | Circular distance | Minimum class distance; distinct from directed interval. |
| SNR009 | Transform | Transposition | T_k(p) = p + k mod n. |
| SNR010 | Transform | Inversion | I_k(p) = k - p mod n. |
| SNR011 | Transform | Retrograde | Reverses an ordered event sequence while preserving event identities. |
| SNR012 | Transform | Mode rotation | Rotates a cyclic scale/motif profile; spelling rules remain profile-bound. |
| SNR013 | Rhythm | Onset | Exact rational or bounded temporal onset. |
| SNR014 | Rhythm | Duration | Exact rational score duration or bounded performance duration. |
| SNR015 | Rhythm | Meter | Measure partition and accent structure. |
| SNR016 | Rhythm | Tempo map | Maps score time to performance time with versioned changes. |
| SNR017 | Rhythm | Tuplet profile | Rational subdivision relation with explicit nesting. |
| SNR018 | Rhythm | Swing profile | Profile-bound timing transform; not a universal interpretation. |
| SNR019 | Event | Note event | Pitch, onset, duration, voice, dynamics and articulation. |
| SNR020 | Event | Rest event | Explicit time-bearing silence; absence of data is not automatically a rest. |
| SNR021 | Event | Chord event | Simultaneous pitch multiset plus voicing and spelling. |
| SNR022 | Event | Control event | Tempo, pedal, controller or parameter event. |
| SNR023 | Event | Articulation event | Typed articulation applied within declared notation/performance profile. |
| SNR024 | Event | Dynamic event | Symbolic or numeric intensity profile with scope. |
| SNR025 | Relation | Voice assignment | Event-to-voice relation with range and polyphony constraints. |
| SNR026 | Relation | Simultaneity | Events share a declared onset region. |
| SNR027 | Relation | Successor | Ordered adjacency in a voice, phrase or global event stream. |
| SNR028 | Relation | Tie | Declared continuation relation; not inferred from identical pitch alone. |
| SNR029 | Relation | Slur/phrase relation | Profile-bound grouping relation, distinct from tie semantics. |
| SNR030 | Relation | Harmonic-function tag | Optional analytical metadata; no cross-cultural universality claim. |
| SNR031 | Guard | Range guard | Pitch/frequency lies in the supported voice/device range. |

| ID | Family | Mechanism | Core contract |
|---|---|---|---|
| SNR032 | Guard | Polyphony guard | Concurrent event count and channel resources are valid. |
| SNR033 | Guard | Quantization guard | Timing/pitch residual lies below a declared margin. |
| SNR034 | Guard | Meter guard | Measure totals and subdivision constraints are satisfied. |
| SNR035 | Guard | Tuning guard | Symbolic spelling/class resolves under the selected tuning profile. |
| SNR036 | Guard | Overlap guard | Voice overlap follows the polyphonic/monophonic policy. |
| SNR037 | Operator | Concatenate | Joins event sequences with explicit time translation. |
| SNR038 | Operator | Overlay | Superposes event sets while preserving voices and conflict policy. |
| SNR039 | Operator | Repeat | References a motif plus transform rather than copying opaque data. |
| SNR040 | Operator | Augment | Scales symbolic durations by an exact factor. |
| SNR041 | Operator | Diminish | Scales symbolic durations by an exact factor. |
| SNR042 | Operator | Voice-leading map | Maps chord voices under range, crossing and cost constraints. |
| SNR043 | Certificate | Symbolic score certificate | Type, meter, range, order, dependency and exact-rhythm scope. |
| SNR044 | Certificate | Performance certificate | Device/profile, timing residual, tuning residual and rendering status. |
| SNR045 | Packing | Motif codebook | Hash-bound local motif/operator aliases with rejection slots. |
| SNR046 | Query | Score graph query | Voice, phrase, interval, motif, dependency and lineage queries. |
| SNR047 | Adapter | MIDI/MusicXML adapter | Versioned interchange mapping; conformance depends on external standards. |
| SNR048 | Projection | Audio render | Downstream synthesis/sample projection; waveform does not redefine score authority. |

# Claims Ledger

| Claim | Disposition | Reason |
|---|---|---|
| T_k and I_k are exact on a declared Z_n pitch-class profile. | ADMIT | They are exact modular transforms inside that selected profile. |
| Twelve-tone equal temperament is universal music. | REJECT | It is one optional tuning profile among many. |
| Enharmonic spellings are always identical. | REJECT | Equivalence depends on tuning, notation, function, voice and historical/cultural profile. |
| Equal pitch class means equal register, voice or identity. | REJECT | Pitch class discards octave/register and does not replace event identity. |
| A mirrored notation glyph automatically gives semantic converse. | REJECT | Converse/inverse semantics must be registered explicitly. |
| A MIDI file is the authoritative musical work. | REJECT | MIDI is an interchange/performance projection with limited semantics. |
| A rendered waveform is the score. | REJECT | Audio is downstream and may vary by synthesis, sample, room and device. |
| Exact rational rhythm can be preserved symbolically. | ADMIT | Onset and duration fractions can remain exact before performance mapping. |
| A seed reconstructs human timing, expression and room acoustics. | REJECT | Those require stored or measured performance evidence. |
| Harmonic-function labels are cross-cultural facts. | REJECT | They are optional analysis metadata bound to a profile. |
| Accessible visual, tactile and auditory routes can share one semantic event graph. | ADMIT WITH PARITY | Meaning, available actions and outcome must remain equivalent across routes. |
| This cartridge proves aesthetic quality. | REJECT | It formalizes structure, execution and evidence, not taste or artistic value. |

# High-Impact Application Profiles

| Profile | What the cartridge enables |
|---|---|
| Executable composition workbench | Drag typed motifs, transforms, meters, voices and guards; preview and commit a deterministic score patch. |
| Music-learning rule world | Pitch classes, intervals, rhythm fractions and operation order become manipulable physical or digital objects. |
| Adaptive game audio | Store motifs and transformation rules instead of opaque stems while keeping gameplay state authoritative. |
| Accessible score editor | One semantic graph supports visual, spoken, tactile, switch, gaze and co-play routes. |
| Performance evidence ledger | Compare score intent with recorded timing/pitch/device residuals without collapsing them. |

# Final Formal Definition

**FORMAL DEFINITION**

UGTOMS SONORA 1.0 is a component-scoped music cartridge in which tuning profiles, pitch/frequency spaces, rational rhythm, meter, voices, score events, transformations, notation OperatorCells, performance evidence and replay lineage are typed substrate data; exact symbolic operations remain distinct from profile-bound analysis and downstream notation, MIDI or audio projections.

## Implementation sequence

7. Freeze JSON Schemas for TuningProfile, ScoreEvent, ScoreGraph, PerformanceEvent and certificates.
8. Implement exact rational time and modular pitch transforms as the scalar oracle.
9. Build an offline browser score editor with safe candidate preview and deterministic replay.
10. Add loss-reporting MIDI and notation adapters behind versioned profiles.
11. Commission diverse musicians and accessibility users before promoting cultural or educational claims.

**RELEASE CONCLUSION**

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.